

DEPARTMENT HANDBOOK

Department of Mechanical Engineering

P.E.S. College of Engineering, Mandya

K V Shankaragowda Rd, PES College Campus

Mandya, Karnataka – 571 401

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Academic Year 2025–26

1. About the Institution

P.E.S. College of Engineering (PESCE), Mandya, was established in 1962 by People's Education Society (PES), Mandya. It is one of the premier engineering institutions in Karnataka, affiliated to Visvesvaraya Technological University (VTU), Belagavi, and approved by AICTE, New Delhi. The college is accredited by NAAC and several of its programmes are accredited by NBA.

The institution has consistently maintained high academic standards and produced graduates who have excelled in industry, academia, and public service across India and abroad.

2. Department Overview

The Department of Mechanical Engineering is one of the founding departments of PESCE and has carved a niche for itself by offering a highly competent instructional programme leading to the B.E. (Mechanical Engineering) degree. The department is also recognised as a Research Centre by VTU and Mysore University, offering both full-time and part-time Ph.D. programmes.

The department is well-equipped with dedicated faculty and focuses on research in design and analysis of machine elements, composite material technology, vibration analysis, and machine vision systems. The quality of teaching and training has yielded a high placement rate at various national and international organisations.

Programme Details

Programme	Type	Duration	Intake
B.E. Mechanical Engineering	UG	4 Years	120
M.Tech. Machine Design	PG	2 Years	As per VTU norms
Ph.D. (Full-time & Part-time)	Doctoral	3–5 Years	Open

3. Vision & Mission

Vision

"Be a department well recognised for its ability to develop competent mechanical engineers capable of working in a global environment."

Mission

- Provide quality education through competent faculty.
- Provide adequate infrastructure and learning ambience for the development of essential technical skills.
- Inculcate a sense of higher education and research orientation.
- Foster industry interaction.

4. Programme Educational Objectives (PEOs)

Graduates of the programme will be able to:

- **PEO 1:** Use the fundamentals of basic science, mathematics, and mechanical engineering to pursue their career as engineers, and to lead and manage teams in global organisations.
- **PEO 2:** Pursue advanced education, research and development, and engage in the process of lifelong learning.
- **PEO 3:** Become entrepreneurs in a responsible, professional, and ethical manner to serve society.

5. Programme Outcomes (POs)

Engineering graduates will be able to:

- **PO 1 – Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.
- **PO 2 – Problem Analysis:** Identify, formulate, and review research literature; analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- **PO 3 – Design/Development of Solutions:** Design creative solutions for complex engineering problems and develop systems, components, or processes meeting identified needs with consideration for public health, safety, and environment.
- **PO 4 – Investigations of Complex Problems:** Conduct investigations using research-based knowledge including design of experiments, modelling, analysis, and interpretation of data to provide valid conclusions.
- **PO 5 – Engineering Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, recognising their limitations, to solve complex engineering problems.
- **PO 6 – The Engineer and the World:** Analyse and evaluate societal and environmental aspects while solving complex engineering problems with respect to sustainability, economy, health, safety, legal framework, and environment.
- **PO 7 – Ethics:** Apply ethical principles; commit to professional ethics, human values, diversity, and inclusion; adhere to national and international laws.
- **PO 8 – Individual and Collaborative Teamwork:** Function effectively as an individual, and as a member or leader in diverse and multi-disciplinary teams.
- **PO 9 – Communication:** Communicate effectively within the engineering community and society, write effective reports, make effective presentations, and comprehend design documentation.
- **PO 10 – Project Management and Finance:** Apply knowledge of engineering management principles and economic decision-making to manage projects in multidisciplinary environments.
- **PO 11 – Life-Long Learning:** Recognise the need for, and have the ability for, independent and lifelong learning, adaptability to emerging technologies, and critical thinking in the context of technological change.

6. Programme Specific Outcomes (PSOs)

- **PSO 1:** Apply computer simulation and experimental methods in the design and development of sustainable products and mechanical systems.
- **PSO 2:** Utilise the knowledge of advanced manufacturing and condition monitoring techniques in industrial applications.

Under the P21 Scheme:

- **PSO 1:** Apply conceptual knowledge with practical engagement to real-life problems by integrating different domains of mechanical engineering.
- **PSO 2:** Utilise modern tools and emerging technologies with technical skills to design, develop, and analyse mechanical systems through a multidisciplinary approach.

7. Teaching Faculty Profiles

Dr. S. Ghanaraja

Designation: Professor

Email: ghanarajas@pesce.ac.in

Educational Qualifications:

- Ph.D. – Nano Composites, IIT Roorkee
- M.Tech. – Maintenance Engineering, SJCE Mysore
- B.E. – Mechanical Engineering, SJCE Mysore

Research Areas: Nano-Composites

Publications: 72 (35 Journals + 37 Conferences)

Dr. Rudresh Addamani

Designation: Professor

Email: rudreshaddamani@gmail.com

Educational Qualifications:

- Ph.D. – Welding, VTU Belagavi
- M.Tech. – Production Management, BVB College of Engineering, Hubballi

Research Areas: Welding, Acoustic Emission, Production/Manufacturing Engineering

Publications: 36 (16 Journals + 20 Conferences)

Dr. Mahendra Babu K J

Designation: Professor

Email: mahendrababukj@pesce.ac.in

Educational Qualifications:

- Ph.D. – Mechanical Engineering Sciences, VTU Belagavi
- M.Tech. – Thermal Engineering, NIT Karnataka

Research Areas: Heat Transfer and Fluid Flow Analysis

Publications: Multiple journals and conferences

Ganapathy Bawge

Designation: Associate Professor

Email: ganapathybawge@pesce.ac.in

Educational Qualifications:

- Ph.D. – Mechanical Engineering Sciences (Super Finishing), PESCE Research Centre, Mandya
- M.Tech. – Maintenance Engineering, SJCE Mysuru

Research Areas: Surface finishing, Corrosion Resistance

Publications: Multiple national and international journals

Dr. Srinivasa M R

Designation: Associate Professor & Head of Department

Email: srinivasamr@pesce.ac.in

Educational Qualifications:

- M.Tech. – Machine Design, BMS College of Engineering, Bengaluru
- B.E. – Mechanical Engineering, PESCE Mandya

Research Areas: Design Engineering, Nano Composite Material, Non-Conventional Machining, NDT

Publications: Multiple journals and conference papers

Dr. S V Anil Kumar

Designation: Associate Professor

Email: svanil999@gmail.com

Educational Qualifications:

- Ph.D. – PSG College of Technology, Coimbatore

Research Areas: Joining Composite Materials

Publications: Several international journal and conference papers

V. C. Chandra Shekara

Designation: Assistant Professor

Email: vccc13@gmail.com

Educational Qualifications:

- M.Tech. – Materials Engineering, NIT Karnataka, Surathkal
- B.E. – Mechanical Engineering, PESCE Mandya

Research Areas: Design Engineering

Publications: —

T M Devegowda

Designation: Assistant Professor

Email: devegowdatm@pesce.ac.in

Educational Qualifications:

- P.G. Diploma in Sugar Engineering, Vasanthdada Sugar Institute, Pune
- B.E. – Mechanical Engineering, PESCE Mandya

Research Areas: Production, Wire EDM and Machine Vision

Publications: Multiple ASME and journal papers

Dr. Mohammad Rafi H. Kerur

Designation: Assistant Professor

Email: mhkerur@pesce.ac.in

Educational Qualifications:

- Ph.D. – Tribology, VTU Belagavi
- M.Tech. – Machine Design
- B.E. – Mechanical Engineering

Research Areas: Tribology

Publications: Multiple national and international papers

Dr. Sadashiva M

Designation: Assistant Professor

Email: sadashivam@pesce.ac.in

Educational Qualifications:

- Ph.D. – Mechanical Engineering, JNTU Anantapur
- M.E. – Manufacturing Science and Engineering, UVCE Bengaluru

Research Areas: Friction Stir Welding, Composite Materials, Manufacturing Science, Advanced Material Technology

Publications: Multiple international journals and conferences; 2 patents published

Dr. Pavan K N

Designation: Assistant Professor

Email: pavankn@pesce.ac.in

Educational Qualifications:

- M.Tech. – Thermal Power Engineering, Nitte Meenakshi Institute of Technology, Bengaluru
- B.E. – Mechanical Engineering, PESCE Mandya

Research Areas: Heat Transfer, Fluid Flow, Nanofluids

Publications: Multiple international journal papers

Dr. Gurupavan H R

Designation: Assistant Professor

Email: gpavan1989@gmail.com

Educational Qualifications:

- Ph.D. – Condition Monitoring, PESCE Mandya
- M.Tech. – Computer Integrated Manufacturing, PESCE Mandya
- B.E. – Mechanical Engineering, Ghousia College of Engineering, Ramanagara

Research Areas: WEDM, Condition Monitoring, Machine Vision

Publications: Multiple ASME and international journal papers

Dr. Rupesh S

Designation: Assistant Professor

Email: mailtorupeshs@gmail.com

Educational Qualifications:

- Ph.D. – Biomass Gasification, NIT Calicut
- M.Tech. – Propulsion Engineering, College of Engineering Trivandrum

Research Areas: Gasification, Bioenergy Conversion

Publications: Multiple Elsevier and Springer journal papers

Dr. Vikram C K

Designation: Assistant Professor

Email: vikramvilas84@gmail.com

Educational Qualifications:

- Ph.D. – Mechanical Engineering, VTU Belagavi
- M.Tech. – Computer Integrated Manufacturing, VTU Belagavi

Research Areas: Computational Fluid Dynamics (Flow past cylinders)

Publications: Multiple ASME and international journal papers

Dr. Raghavendra S

Designation: Assistant Professor

Email: raghavendras@pesce.ac.in

Educational Qualifications:

- Ph.D. – Composite Materials, VTU Belagavi
- M.Tech. – Computer Integrated Manufacturing, PESCE Mandya
- B.E. – Mechanical Engineering, PESCE Mandya

Research Areas: Composite Materials

Publications: —

Syed Imran Ali

Designation: Assistant Professor

Email: syedimranali@pesce.ac.in

Educational Qualifications:

- M.Tech. – CIM, PESCE Mandya
- B.E. – Industrial and Production Engineering, PESCE Mandya

Research Areas: Production and Manufacturing

Publications: —

Somashekar B R

Designation: Assistant Professor

Email: vijayvijay.oct20@gmail.com

Educational Qualifications:

- M.Tech. – Thermal Power Engineering, Alva's Institute of Engineering & Technology, Moodbidri
- B.E. – Mechanical Engineering, BTL Institute of Technology & Management, Bengaluru

Research Areas: Thermal Engineering

Publications: —

Doddaswamy V

Designation: Assistant Professor

Email: doddaswamyv@pesce.ac.in

Educational Qualifications:

- M.Tech. – Machine Design, PESCE Mandya
 - B.E. – Industrial and Production Engineering, PESCE Mandya
- Research Areas:** Fatigue, Fracture and Vibration of Metal Matrix Composites

Publications: Multiple international journal papers

Pavan Krishna K

Designation: Assistant Professor

Email: pavankrishnak@pesce.ac.in

Educational Qualifications:

- M.Tech. – Raja Rajeshwari College of Engineering, Bengaluru
- B.E. – Vidya Vardhaka College of Engineering, Mysuru

Research Areas: Fuel Cells, CFD, Heat Transfer

Publications: —

Dr. Lakshmi Narasimha Murthy H R

Designation: Assistant Professor

Email: narsimhr@gmail.com

Educational Qualifications:

- Ph.D. – Mechanical Engineering, VTU Belagavi
- M.Tech. – Maintenance Engineering, SJCE Mysore

Research Areas: Composite Materials, Machinability Optimisation

Publications: Multiple international journal papers

Avinash M

Designation: Assistant Professor

Email: mandyaavinash@gmail.com

Educational Qualifications:

- M.Tech. – Computer Integrated Manufacturing, PESCE Mandya
- B.E. – Mechanical Engineering, Ghousia College of Engineering, Ramanagara

Research Areas: Computer Integrated Engineering

Publications: International journal paper on Friction Stir Welding

Dr. Srikanth Shekar K C

Designation: Assistant Professor

Email: shekarsrikanthkc@gmail.com

Educational Qualifications:

- M.Tech. – VTU Regional Centre, Mysuru
- B.E. – PESCE Mandya

Research Areas: Thermal Engineering

Publications: —

Anand K

Designation: Assistant Professor

Email: anand08.k@gmail.com

Educational Qualifications:

- M.Tech. – Thermal Power Engineering, VTU Regional Centre, Mysuru
- B.E. – Mechanical Engineering, Ghousia College of Engineering, Ramanagaram

Research Areas: Thermal Engineering

Publications: —

8. Technical / Non-Teaching Staff

Name of Staff	Designation	Qualification	Experience (Yrs.)
H.S. Chandra Shekar	Foreman	DME	32
B. Sathish Chandran	Instructor	B.E.	31
Hanumantha Shetty	Instructor	DME	30
J. Ramesh	Asst. Instructor	DME	14
M.Y. Manjesh	Asst. Instructor	DME	14

K. Tharakeshwara	Asst. Instructor	DME	12
Devaraju	Asst. Instructor	DME	12
Madegowda H	Asst. Instructor	DME	12
Ravi B	Asst. Instructor	DME	12
A.M. Raju	Asst. Instructor	MA	26
H.S. Charan	Mechanic	ITI	14
C. Nagaraju	Mechanic	ITI	14
R.N. Mallesha	Mechanic	DME	14
Sharath S L	Mechanic	ITI	05
M. Shiva Kumar	Mechanic	SSLC	21
M.M. Mahesh	Helper	SSLC	30
K. Gopal Raju	Helper	SSLC	21
K.S. Kalyan Kumar	Helper	JOC	21
Kendiahswamy	Helper	SSLC	14
S.S Varadharaju	Helper	PUC	14
H Parvathi	Helper	ITI	09
Nagesh S. J.	Helper	PUC	09
K.N. Armugam	Helper	SSLC	33
M N Manjunatha	Helper	SSLC	36

9. Research & Development

Research Guides

Research Guide	Phone Number	Email
Dr. Ghanaraja S	+91 9449149167	ghanaraja08@gmail.com
Dr. Rudresh Addamani	+91 9739102611	rudreshaddamani@gmail.com
Dr. K.J. Mahendra Babu	+91 9448754666	babukjm1008@yahoo.com
Dr. S V Anil Kumar	+91 9448799499	svanil999@gmail.com
Dr. Sadashiva M	+91 9972212773	sadashiva015@gmail.com
Dr. Gurupavan H.R	+91 9686632650	gpavan1989@gmail.com
Dr. Rupesh S	+91 9846048651	mailtorupeshs@gmail.com
Dr. Madhusudana C K	+91 7338513801	madhusudanack.008@gmail.com

Ph.D. Scholars Awarded

Scholar Name	Guide	Title of Research	Date of Viva
Mohammad Rafi H Kerur	Dr. T. Nagaraju	3D Surface Roughness on THD Analysis of Porous Bearing	July 2021
Naveed Anjum	Dr S L Ajit Prasad	Processing and Characterisation of Particulate Filled and Natural Fiber Reinforced Polymer Composites	June 2020

Publications Summary

Faculty Name	International/National Journals	International/National Conferences	Total
Dr. S. Ghanaraja	35	37	72
Dr. Rudresh Addamani	16	20	36

Patents

App. No.	Date	Title	Faculty	Status
216476	2003	An Extrusion Honing Composition	Dr. H.P Raju	Granted
1172/Del/2010	19/05/2010	Processing of Alumina Nano-Particle Reinforced Cast MMC	Dr. S. Ghanaraja	Applied
201841026027	2018	Single Wheeled Self-Balancing Electric Vehicle	Dr. C.J. Gangadhara Gowda & Shreyas G.V	Granted
2021102713	20/05/2021	Fully Autonomous Smart Manufacturing Unit	Dr. Sadashiva M	Published
202141032004	16/07/2021	Online Reinforced Learning of Assembly Sequence Planning	Dr. Sadashiva M	Published
202041050076	27/11/2020	Intelligent Car Parking System Using IoT	Dr. Anil Kumar S V	Published
202141043144	01/10/2021	Novel Rotary Flow Control Valve in Gas Turbine Fuel Injection	Dr. K J Mahendra Babu	Published
202141043279	05/11/2021	Solar Water Boiling System with Thermoelectric Capacity	Mr. Siddesh Kumar N M	Published
202141043085	05/11/2021	Cheap Verbalized Robotic Hand for Specific Adherence	Mr. Ramesh Kurbet	Published

10. NBA Accreditation

Programme Name	Duration	Accreditation Start	Accreditation End
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Mechanical Engineering	3 Years	2023	2026
Mechanical Engineering	3 Years	2020	2023
Mechanical Engineering	3 Years	2017	2020

11. Results & Roll of Honour

Programme Results

Year of Pass	Students Enrolled	Students Graduated
2022-23	165	161
2023-24	115	112
2024-25	135	132

Roll of Honour – Academic Year 2023-24

Student Name	USN	CGPA	Rank
Dheeraj N U	4PS19ME041	8.75	First
Sushmitha S	4PS19ME116	8.53	Second
Mahesh M	4PS20ME414	8.49	Third

Roll of Honour – Academic Year 2022-23

Student Name	USN	CGPA	Rank
Chandan N	4PS18ME016	9.09	First
Santhosh R	4PS18ME077	9.01	Second
Darshan N	4PS18ME022	8.95	Third

12. Distinguished Alumni

- **Dr. B.N. Suresh** (Batch of 1967): Padma Shri (2002) and Padma Bhushan (2013) awardee. Renowned aerospace engineer with contributions in space technology, launch vehicle design, aerospace navigation, control and actuation systems.
- **Mr. Girish Bharadwaj** (Batch of 1973): Padma Shri awardee. Constructed around 127 hanging bridges across streams and rivers in remote areas of Karnataka, bringing development to far-flung communities.

- **Mr. Kanumuru Satyanarayana Raju:** Chairman of the Nagarjuna Group (Agri Business, Power, and Refining), the largest industry conglomerate in Andhra Pradesh.
- **Dr. S. Guruprasad** (Batch of 1983): Key contributor to DRDO R&D activities; member of the design team for Project Sarvatra, a multi-span Mechanical Military Bridging System.
- **Dr. B.R. Ravikanthe Gowda IPS:** Joint Commissioner of Police, Traffic, Bangalore City.
- **Mr. Girish:** CEO, Test Yantra Software.
- **Prof. Sridar T S** (Batch of 1973): Retired Director and Vice Chancellor, Jain University.
- **Mr. B N Vasudev** (Batch of 1972): Retired General Manager, Karnataka Small Industries Development Corporation.
- **Mr. Gurumurthy** (Batch of 1971): GM in a Swiss Calibrated Company engaged in manufacturing high-voltage electrical insulation.
- **Flt. Lt. K P Nagesh** (Batch of 1986): MD, UniQ Group of Companies, Bangalore.
- **Mr. I.J. Umakanth** (Batch of 1970): Retired GM, Forbes Group.
- **Mr. Rabindra Simon** (Batch of 1967): Retired Chief Engineer (C&B and Irrigation), PWD, Government of Karnataka.
- **Mr. G B Shivakumar** (Batch of 1973): Ex-MLA, Keragodu Constituency; Ex-MLC, Mandya.
- **Dr. J. Venkateshwaram** (Batch of 1970): Ex-Scientist, NASA, USA.

13. Departmental Activities

The department regularly organises a variety of technical, co-curricular, and extracurricular activities to enhance student learning and holistic development. Key activities include:

- **Standard Club Inauguration (BIS) (04 July 2023):** Inauguration of the Bureau of Indian Standards club to promote standardisation awareness.
- **Hand Layup Techniques Workshop (07 February 2023):** Hands-on workshop on composite manufacturing using hand layup processes.
- **Boot Camp (19-20 January 2023):** Intensive skill development camp for mechanical engineering students.
- **Mecho Cup (16-27 January 2023):** Inter-collegiate technical competition promoting engineering innovation.
- **Technical Talk – CFD in Aerospace (14 January 2023):** Expert lecture on basics and applications of Computational Fluid Dynamics in aerospace industries.
- **9th GKDC (22-25 August 2022):** Participation in the Go-Kart Design Challenge competition.
- **Srishti State Level Exhibition (2021-22):** Secured Second Prize at the Srishti state-level technical exhibition.
- **Smart India Hackathon (2021-22):** Department team reached the finals of the Smart India Hackathon.
- **Orientation and Induction Programmes (Annually):** Orientation and induction programmes conducted for new students every academic year.

14. Contact Information

Institution	P.E.S. College of Engineering, Mandya
Department	Mechanical Engineering
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Phone	08232 220043
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Website	www.pesce.ac.in
HOD	Dr. Srinivasa M R, Associate Professor & Head of Department
HOD Email	srinivasamr@pesce.ac.in